

ADEN GUO

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TECHNICAL SKILLS

- **Languages & Data:** Python, TypeScript, JavaScript, Go, Rust, Ruby, C++, SQL, NumPy, Pandas
- **AI & LLM Engineering:** Anthropic / Claude API, OpenAI, Microsoft Foundry, LangChain, LangGraph, Vercel AI SDK, OpenRouter, RAG & hybrid vector retrieval, Embeddings, Prompt caching, System-prompt design
- **Agentic Development:** MCP servers, Claude Code & Codex CLI, Agent skills & subagents, Repo-scoped agent context (AGENTS.md), Hook-driven automation, n8n workflow automation, Persistent / long-running agent runtimes
- **Backend & Data:** FastAPI, Node.js, Ruby on Rails, REST API design, PostgreSQL, pgvector, SQLite, Redis, Drizzle ORM, Supabase, Neon
- **Web & Frameworks:** React, Next.js (App Router), SvelteKit / Svelte 5, Tailwind CSS, Tauri, PWA & service workers, Bun
- **Cloud & DevOps:** Azure (App Service, Functions, API Management), Azure DevOps (Repos & Pipelines), Docker, CI/CD, Vercel, Railway, AWS S3, Linux, Bash, Git, Playwright, pytest / RSpec
- **Geospatial:** ArcGIS Pro & Online, QGIS, PostGIS, GeoPandas, GDAL/PROJ, FME Form & Flow, MapLibre GL, RTK GNSS

PROFESSIONAL EXPERIENCE

GIS Technologist - Canada West Land, Calgary, AB

Dec 2025 - Present

- Deploy internal-facing APIs on Azure — App Service web apps, Azure Functions for event- and schedule-triggered jobs, and API Management for gateway routing, auth, and rate limiting — shipped via Azure DevOps with Repos code review, Pipelines CI/CD, and Doppler-managed secrets.
- Designed and deployed role-specific internal agents on Microsoft Foundry, grounded on internal company data and wired to internal APIs and tool actions; now in daily production use by colleagues.
- Contributed to an internal Azure-hosted production app for agentic AI workflows using LangChain, LangGraph, and n8n with role-specific system-prompt design, cutting selected full-day manual processes to 1–2 hours.
- Build and maintain internal geospatial tooling through agentic CLI development (Claude Code, Codex) with repo-scoped agent context and hook automation, replacing manual ETL, QC, and map-production steps; several of these methods are now standard on the team. Prototyped self-hosted long-running agent runtimes with cross-session state and remote triggers.
- Key contributor on a large-scale enterprise GIS delivery for a major client: own a six-map production workstream and its end-to-end ETL and data-cleaning pipelines (Python, Esri APIs, FME) across datasets of roughly 5,000 to 1M records.

Engineering Capstone Project Lead - University of Calgary, Calgary, AB

Sep 2024 - Apr 2025

- Led capstone delivery for a geospatial mapping study, building the Python processing and validation pipeline used to compare survey methods against photogrammetry-derived spatial datasets.

Shift Lead / Team Trainer - Gongcha

Nov 2020 - May 2024

- Managed high-volume shift operations and trained staff on standardized workflows, strengthening team coordination and execution under time constraints.

EDUCATION

Bachelor of Science: Geomatics Engineering

University of Calgary - Calgary, AB

2020 - 2026

PROJECTS

Glass Atlas: Source-Grounded RAG Knowledge Site - glass-atlas-production.up.railway.app

- Deployed SvelteKit 2 (Svelte 5 runes) site that turns a personal note library into a conversational, citation-backed assistant.
- Engineered hybrid semantic + lexical retrieval over Neon PostgreSQL with pgvector and Drizzle ORM, with confidence-tiered fallbacks that keep every answer grounded in source notes.
- Built streaming RAG chat through OpenRouter (Gemini 2.0 Flash) on 1536-dim text-embedding-3-small vectors, strictly typed end to end in TypeScript.
- Shipped on Railway with a Bun HTTP server, Auth.js GitHub OAuth, and private S3-compatible storage via presigned URLs.

PigeonCoop: Local-First Agent Workflow Engine - github.com/AG9898/PigeonCoop

- Rust/Tauri desktop app for designing, running, monitoring, and replaying coding-agent workflows against local repositories.
- Designed Rust core crates for typed workflow definitions, RunEvent envelopes, deterministic run/node state machines, and SQLite-backed workflow, run, and event persistence.
- Built runtime adapters that drive real coding-agent CLIs, plus a React/TypeScript graph workspace with Agent, Tool, Router, Memory, and Human Review nodes and pre-run validation.
- Covered builder, live-run, human-review, and replay flows with Rust unit tests, React component tests, and WebdriverIO/Tauri end-to-end suites.

Weather & Wellness: UBC Research Platform - ubcpsych.com

- Built a Next.js 15 / React 19 frontend and FastAPI backend that auto-scores validated instruments (digit span, loneliness, depression, anxiety) in real time.
- Architected per-lab data isolation on Supabase Postgres/Auth with Upstash Redis, and mixed-effects weather-psychology modelling via statsmodels.